

LinearLLMDynamic — Remaining Experiments (Handoff)

Status as of 2026-09-05 · repo: LinearLLMDynamic (branch main) · contact figures: AppliedController/results_reports/final_summary_*.png

The project produces **6 summary figures** comparing 4 methods (Original, A-LQR, S-PID, H-infinity) across model scales on truthfulness, in-distribution toxicity (RealToxicityPrompts) and 3 out-of-distribution toxicity benchmarks. Models whose data is only **smoke-level** (reduced samples, provenance run_status=smoke_test) or missing are shown with **gray labels** in every figure; fully-run models are black. The table below lists what remains to make every label black.

#	Experiment	Model	Current status	What's needed	Priority
1	Truthfulness (paper-aligned, full)	Qwen-2.5-0.5B	Smoke only (100 TruthfulQA / 50 MMLU samples)	Full run: all TruthfulQA + full MMLU 5-shot, quantized. Must pass --device cuda:0 (bare 'cuda' crashes the device parser).	High
2	RTP toxicity (calibrated, full)	Qwen-2.5-0.5B	Smoke only (25 prompts)	Full-prompt calibrated steering run, then re-export paper_style_table_paper_like_calibrated.csv (replace smoke rows).	High
3	OOD global-3 (full)	DistilGPT-2	Smoke only (25 prompts per subset)	Full run of jigsaw_long, toxicchat_long, mmlu_ood_other_concepts x 3 controllers. Export without the _smoke suffix.	High
4	OOD global-3 (full)	Qwen-2.5-0.5B	Smoke only (25 prompts per subset)	Same as #3, using experiment_plan_ood_global3_qwen05b.json.	High
5	RTP toxicity	Qwen-2.5-1.5B	Never run (N/A row in toxicity figure)	Calibrated steering run on RealToxicityPrompts; fills the gray 1.5B row.	Medium
6	OOD global-3	Qwen-2.5-1.5B	Never run (N/A rows in OOD figures)	Steering run on the 3 global OOD subsets; fills the gray 1.5B rows.	Medium
7	Gain / lambda sweep (A-LQR, H-infinity)	Qwen-2.5-0.5B (cheapest)	Deferred by project owner	Sweep controller gains to fix over-steering collapse (True ~95%, Info ~1%) confirmed on 0.5B and 1.5B. Unblocks better truthfulness numbers for A-LQR / H-infinity everywhere.	Medium
8	Truthfulness A-LQR / H-infinity	Qwen-2.5-7B	N/A	Excluded by project owner decision (no more 7B/14B runs). Rows stay as red N/A markers.	Off
9	All toxicity / OOD	Qwen-2.5-7B / 14B	N/A	Excluded by project owner decision.	Off

How to run the remaining experiments

0. Environment

Lightning.ai studio, single H100 80GB. Interpreter: /teamspace/studios/this_studio/.venv/bin/python. Always run modules from the repo root:

```
cd /teamspace/studios/this_studio/LinearLLMDynamic
/teamspace/studios/this_studio/.venv/bin/python -u -m AppliedController.<module> ...
```

Critical gotcha: quantized model loading parses the device string as `device.split(':')[1]` — always pass **--device cuda:0**, never bare 'cuda' (IndexError otherwise). Push every commit to BOTH remotes: origin (erfanzabeh/LinearLLMDynamic) and upstream (rgs2151/controller).

1. Truthfulness full run (experiments #1)

Same command as the smoke run in `results_reports/truthfulness_rerun_runs/qwen05b_smoke.log`, but drop `--truthful-samples / --mmlu-samples` limits:

```
python -u -m AppliedController.eval_truthfulness_paper_aligned \
--model Qwen/Qwen2.5-0.5B --quantized --device cuda:0
```

Then merge the output CSV into `results_reports/paper_style_table_truthfulness_all_models_methods.csv` (model key 'Qwen-2.5-0.5B-smoke' → rename to 'Qwen-2.5-0.5B-ours', set `run_status=full`, update `TRUTH_MODEL_ORDER/LABELS` in `figure_config.py`).

2. Steering runs: RTP + OOD full (experiments #2-6)

Experiment plans already exist: `AppliedController/experiment_plan_ood_global3.json` (DistilGPT-2) and `experiment_plan_ood_global3_qwen05b.json` (0.5B; includes `rtp_random`). The smoke runs used `--eval-prompts 25 --limit-experiments 4` — remove both flags for full runs:

```
python -u -m AppliedController.run_steering \
--experiment-plan AppliedController/experiment_plan_ood_global3_qwen05b.json \
--device cuda:0 --output results_reports/ood_runs/steering_results_ood_global3_qwen05b_full
```

For Qwen-2.5-1.5B (#5, #6): create a new plan JSON by copying the 0.5B plan and swapping `model_name` to `Qwen/Qwen2.5-1.5B`, label to 'Qwen-2.5-1.5B'. Then export the paper-style table:

```
python -u -m AppliedController.export_ood_paper_style \
--input results_reports/ood_runs/steering_results_..._full.csv \
--output results_reports/paper_style_table_ood_global3.csv --provenance full
```

Toxicity metric: continuation toxicity % scored with `s-nlp/roberta_toxicity_classifier`. The 3 global OOD subsets are fixed across all models (chosen from residual analysis): `jigsaw_long`, `toxicchat_long`, `mmlu_ood_other_concepts`.

3. Gain / lambda sweep (experiment #7)

Both A-LQR and H-infinity over-steer on truthfulness (True ≈ 95%, Info ≈ 1%, so T·I collapses) — confirmed on 0.5B and 1.5B; S-PID does not collapse. Sweep the controller gain / lambda on the cheap 0.5B model, pick the value maximizing T·I without MMLU degradation, then re-run truthfulness for affected models. This was explicitly deferred by the project owner — confirm before spending GPU time.

4. After any full run: update figures

a) Merge new rows into the relevant CSV (keep run_status provenance column).

b) In AppliedControler/figure_config.py add the model to TRUTH_FULLY_RUN / TOX_FULLY_RUN / OOD_FULLY_RUN so its label turns black. Layouts are append-only: add models/rows only via figure_config, never restructure figure scripts.

c) Re-render all six figures:

```
for s in make_final_summary_figure make_final_summary_model_dissected \
    make_final_summary_model_dissected_toxicity make_final_summary_figure_ood \
    make_final_summary_model_dissected_ood; do python -u -m AppliedControler.$s; done
```

Figure map (all in AppliedControler/results_reports/)

Figure stem	Content	Data source
final_summary_figure	Truthfulness summary (bar, models on x) + DistilGPT-2 toxicity panel	truthfulness CSV + calibrated toxicity CSV
final_summary_model_dissected	Truthfulness dissected: rows=models, cols=T-I/True/Info/MMLU	paper_style_table_truthfulness_all_models_methods.csv
final_summary_model_dissected_toxicity	Toxicity dissected: rows=models, cols=rate(log)/reduction	paper_style_table_paper_like_calibrated.csv
final_summary_figure_ood[_smoke]	OOD summary: 1x3 panels (one per OOD subset), models on x	paper_style_table_ood_global3_smoke.csv
final_summary_model_dissected_ood[_smoke]	OOD dissected: rows=models, cols=3 subsets, toxicity % (log)	same OOD table
final_summary_model_dissected_ood_reduction[_smoke]	OOD reduction dissected: same grid, -percent_change (linear)	same OOD table

[_smoke] suffix is applied automatically when the input table filename contains 'smoke'; it disappears once a full table is used.

Key findings so far (context for interpreting new runs)

- **S-PID** is the only controller that improves truthfulness without collapsing informativeness (0.5B smoke: T-I 46.2 vs Original 43.3; same pattern at 1.5B).
- **A-LQR / H-infinity over-steer** on truthfulness at current gains (Info $\approx$ 1%) — motivates experiment #7.
- **In-distribution toxicity:** DistilGPT-2 full run: large reductions all methods. Qwen-0.5B smoke: baseline 14.9% $\rightarrow$ A-LQR 6.43 (–56.7%), H-inf 9.88 (–33.5%), S-PID 14.1 (–5.3%) — A-LQR/S-PID ranking flips vs DistilGPT-2.
- **OOD transfer:** DistilGPT-2 controllers are flat on jigsaw_long (baseline 67.9%, $\pm$ 0.4%) — rtp-calibrated gains do not transfer; genuine finding, not a bug. Qwen-0.5B shows modest real transfer (72.7 $\rightarrow$ 67.3 A-LQR). toxicchat_long / mmlu_ood are near the classifier floor ($\sim$ 0.01%) for small models and are weakly discriminating.
- 7B/14B rows are intentionally N/A (owner decision: no more large-model runs).